用 Python 探索數據與預測

Explore data and predict with Python

班別 / Class: _____ 組別 / Group: _____ 日期 / Date: _____

先儲存 Colab 副本。每節按一次程式格左邊的播放按鈕，等待結果。

Save a Colab copy. Press the play button once per section and wait for results.

以 # 開頭的行是說明用的註解。無需背誦程式或手算回歸。

A line starting with # is a comment. No memorising code or hand calculation is needed.

1 數據探索：外送距離與收費

Data exploration: delivery distance and fee

這是模擬紀錄；km 是距離（公里），fee 是收費（港元）。

Simulated records: km is distance (km); fee is delivery fee (HK$).

兩個列表相同位置是一對數據，例如 1 公里、24 元；每點是一筆紀錄。

Matching positions form a pair, e.g. 1 km and HK$24. Each dot is one record.

```
# 載入畫圖工具 / Load the plotting tool
import matplotlib.pyplot as plt

# 每對數值是一筆外送紀錄 / Each pair is one delivery
km = [1, 2, 3, 4, 5, 6, 7, 8]
fee = [24, 29, 27, 38, 35, 44, 42, 51]

# 畫散點圖；每點是一筆紀錄 / Plot one dot per record
plt.figure(figsize=(8, 4))
plt.scatter(km, fee, color="#086c61", s=65)
# 標示座標軸及單位 / Label the axes and units
plt.xlabel("Delivery distance (km)")
plt.ylabel("Delivery fee (HK$)")
plt.title("Simulated delivery data")
plt.xlim(0, 9)
plt.ylim(0, 60)
plt.grid(alpha=0.2)
plt.tight_layout()
plt.show()
```

1. 距離增加時，收費大致怎樣變化？

As distance increases, how does the fee generally change?

2. 距離較遠，收費是否每次都較高？用兩筆紀錄說明。

Is the fee always higher for a longer distance? Use two records to explain.

資源 / Resources: gordany.github.io/f2-linear-regression/workshop-2/

2 線性回歸：每天溫習時間與成績
Linear regression: daily study time and exam score

每對數據是一位模擬學生。用直線概括趨勢，再預測每天 4.5 小時。

Each pair is a simulated student. Fit a trend line, then predict for 4.5 h/day.

m（變化率）：每多一小時，估計分數的改變；c（初始值）：0 小時的估計分數。

m: change in estimated marks per extra hour. c: estimated marks at 0 hours.

```
# 載入畫圖及模型工具 / Load plotting and model tools
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# 每對數值是一位模擬學生 / Each pair is a simulated student
hours = [0, 1, 2, 3, 4, 5, 6]
scores = [53, 46, 59, 52, 63, 54, 65]
# 將時間排成模型需要的格式 / Format inputs for the model
X = [[h] for h in hours]
# fit 從數據找最佳擬合直線 / fit finds the best-fit line
model = LinearRegression().fit(X, scores)
m = model.coef_[0]
c = model.intercept_
# 印出模型摘要 / Print the model summary
print("Model summary: observations =", len(hours))
print(f"Line: y = {m:.2f}x + {c:.2f}")
print(f"m (rate of change): {m:.2f} marks per extra hour")
print(f"c (initial value): {c:.2f} marks at 0 hours")
# predict 用模型估計新輸入 / predict estimates a new input
study_hours = 4.5
prediction = model.predict([[study_hours]])[0]
print(f"Prediction at {study_hours} h/day: {prediction:.1f} marks")

# 同圖顯示數據、直線和預測 / Plot data, line and prediction
plt.figure(figsize=(8, 4))
plt.scatter(hours, scores, color="#086c61", label="Data")
plt.plot([0, 6], model.predict([[0], [6]]),
        color="#b65320", label="Best-fit line")
plt.scatter(study_hours, prediction, color="#6b51a3",
           marker="D", s=85, label="Prediction")
plt.annotate(f"{study_hours} h: {prediction:.1f} marks",
            (study_hours, prediction), xytext=(8, 78),
            arrowprops=dict(arrowstyle="->", color="#6b51a3"))
# 保留完整時間及分數範圍 / Show the full time and score ranges
plt.xlim(0, 24)
plt.ylim(0, 100)
plt.xlabel("Daily study time (hours)")
plt.ylabel("Exam score (marks)")
plt.legend()
plt.grid(alpha=0.2)
plt.tight_layout()
plt.show()
```

1. 圖中每天 4.5 小時的預測成績是 _____ 分。

The predicted score at 4.5 hours per day is _____ marks.

2. 最佳擬合直線是否通過每一點？寫下你的觀察。

Does the best-fit line pass through every dot? Write your observation.

預測是估計，並非保證；模擬數據不能證明因果。

A prediction is an estimate, not a guarantee; simulated data do not prove cause.